

# Quality Control & Determining the Acceptance Threshold of Scintillating Tiles for Super MuSR

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## Abstract

This report aims to determine a acceptance threshold for novel 30mm scintillating tiles for Super MuSR and the associated percentage of tiles that are approved for the assembly stage. 2.7% of tiles break before testing and 4.9% of tiles that are tested do not have suitable light output. Therefore 92.5% of all 30mm tiles produced are expected to be approved for the detector. Pulse Height Spectra (PHS) peak Values were extracted to determine light output, and compared to a benchmark tile with mean PHS peak of  $0.2462 \pm 0.0820\text{V}$ . The batch of 41 tiles tested in this report that had a mean PHS peak of  $0.2978 \pm 0.0216\text{V}$ , a 20.9% improvement. The testing procedure used a Photomultiplier Tube setup and a collimated  $^{90}\text{Sr}$  source to simulate positrons. Bulk transmittance of epoxy resins used in the tiles is also investigated, determining that IN2 performs better RAL71a resin in terms of this metric.

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## 1 Introduction

The Detector Systems Group (DSG) is the team at the ISIS Muon and Neutron Source (ISIS), responsible for development, electronics, assembly and maintenance of the array of current and

future detectors at ISIS. Work on these detectors is crucial for the advancement of radiation detection technology and has extensive applications in both academia and industry [1]. One of the detectors is MuSR, a Muon Spectrometer primarily focussing on research in magnetism and superconductivity using Muon Spin Rotation/Relaxation/Resonance. It has 64 detector elements, a temperature range of 40mK-1000K, and fields of 0-300mT [2]. Super MuSR is the upgrade for MuSR which aims improve count rate by a factor of 20, in addition to a factor of 6-10 improvement in time resolution. There will also be an increase of the main magnet, going from 0-650mT, in addition to a higher transverse field available through the use of new spin rotators in the beam line. The overall purpose of this is to maximise data per pulse [3]. ISIS gets visits from approximately 3500 scientists yearly from across the globe, and MuSR was one of the most over-subscribed instruments, with higher rate of data collection, Super MuSR can increase the experimental capacity, driving advancements across multiple different fields [4]. The applications of Super MuSR are vast including research into complex materials, liquid crystal dynamics, quantum technologies, battery technology, and further applications in life sciences [5]. This report focusses the development, characterisation and quality control of novel scintillating tiles for Super MuSR.

## 2 Theory

### 2.1 MuSR Fundamentals

The Start of a Muon Experiment begins with the production of Muons. This is done using the Synchrotron at ISIS. To begin,  $H^-$  ions are produced in an ion source, and extracted in 200 $\mu$ s long injections of ions. This forms the base particles that will be accelerated and will arrive at the detectors in pulses separated in time by 4.94ns. These pulses are injected and accelerated using a Radio Frequency Quadrupole. These are passed into a linac, which accelerates the beam to 70MeV, this is the point where the ion bunches are stripped of their electrons by a thin aluminium film, where the the now protons are fed into the synchrotron. Using a series of bending and accelerating magnets the synchrotron accelerates the protons to 800MeV. The protons are then sent down the beamline to one of two target stations. Target Station 1 (TS1) receives 4 pulses of protons, and is responsible for higher rate and energy experiments, compared to TS2 which receives the remaining pulse for optimised experiments involving lower energy and longer wavelength neutrons. Both Target Stations produce neutrons via spallation [7]. Upstream of the TS1 spallation source is a thin carbon target, which produces pions which decay into muons. The muon group at ISIS produce both positive and negative muons, but MuSR only receives positive muons which are produced with the following nuclear reactions [8]:

$$p + p \rightarrow p + n + \pi^+ \quad (1)$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu \quad (2)$$

There are a specific "type" of muons that are of interest, called surface muons. These are when the pions come to rest near the surface of the carbon target. When they decay, they produce polarised muons due to the fact that pion decay violates parity, so the muon spin must align with its own momentum. This is important as it allows the determination of the initial momentum of the muons and allows the accurate measurement of spin-momentum correlation [9]. These surface muons are accelerated down the muon arm of TS1, which functions as a velocity selector ensuring only the muons with high levels of polarisation reach the sample. Now that the muons are implanted in the sample, muon Spin Rotation/Relaxation/Resonance can be conducted. This is the process used to measure the local magnetic field of a sample. It does this by measuring the decay of a muon that has been implanted into a sample, and beings Larmor Precession (figure 2) where the spin direction of the muon "wobbles" about the direction of the magnetic field it is in. The Lifetime of a Muon is 2.2 $\mu$ s and for a given pulse of muons arriving at the sample at the same time the decay is governed by:

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Figure 1: Schematic of the experimental Halls and Synchrotron of ISIS [6]. Note the muon beamlines arms Highlighted in purple, coming off TS1.

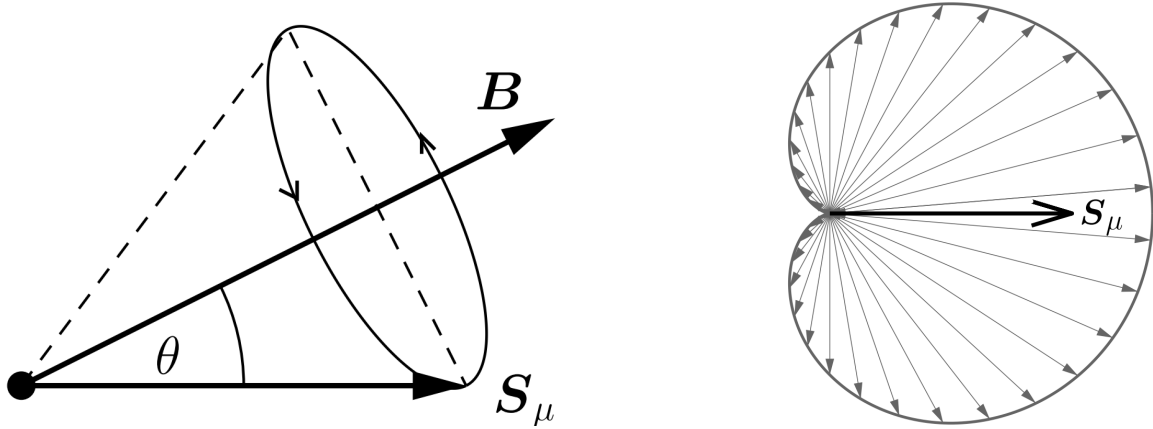


Figure 2: A visualisation of Larmor precession (left) and the probability distribution of muon decaying to a positron (right) [8]

$$n(t) = n(0)e^{-t/\tau_\mu} \quad (3)$$

where  $\tau_\mu$  is the average muon lifetime. Each individual muon particle decays into a positron and neutrinos as follows:

$$\mu^+ = e^+ + \bar{\nu}_\mu + \nu_e \quad (4)$$

The decay of a muon is asymmetric (figure 2). This means there is more counts, on average, are greater in the direction of the magnetic field of the sample the muon is implanted within, this allows the magnetic composition of various samples to be determined. [8].

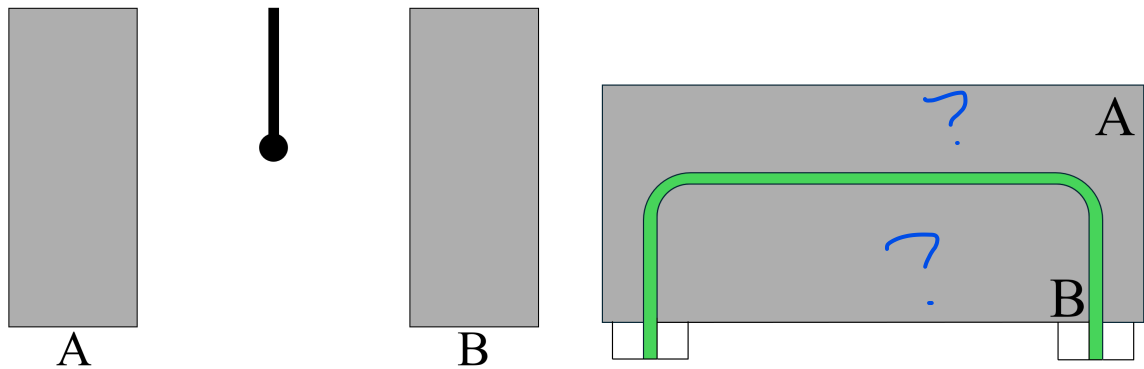
## 2.2 Super MuSR Design

illustration?

MuSR uses scintillator and wavelength shifting fibre that directs light to an array of 64 Photomultiplier Tubes (PMT). A PMT works by producing photoelectrons from the photo-cathode, when struck by light produced by the scintillator, which are accelerated through a variety of high voltage dynodes, multiplying the number of photoelectrons. This electrical signal is generated and converted into a pulse on an oscilloscope, determining the intensity of light emitted from the scintillator onto the photo cathode, and subsequent energy of the incident radiation on the scintillator can be determined [10]. Super MuSR intends to use an array of 960 individual detectors. Each of these detectors is made up of one scintillating tile, attached to two Silicon Photomultipliers (SiPMs). These are a solid-state photodetectors with gain similar to that of a Photomultiplier Tube. They are made of an array of pixelated, Geiger mode photodetectors. Each pixel on an SiPM can register as either triggered or not triggered. As a result the light entering the SiPM will trigger a discrete number of pixels, which then corresponds to a voltage and light output [11]. To focus light to the SiPMs an array of scintillating tiles at varying lengths and thickness are arranged in a barrel shape in the detector. The tiles are made of a plastic scintillator, a wavelength shifting fibre (WLSF) to direct light to the SiPMs, and a clear epoxy resin. Each tile is wrapped with "Excellent Spectral Reflector" (ESR) to prevent light escaping one produced by scintillation. At each end of the fibre there is a stud, which is polished down and allows the WLSF to fit over the SiPMs with a small tolerance, making an effective and compact positron detection system. R&D and Quality control of these tiles is imperative for a successful detector, improving light output and energy resolution.

are there any specific targets for these?

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(a) A arbitrary sample being held in the centre of two detectors for a standard MuSR experiment, if detector A were to have a higher count rate compared to detector B, the magnetic field of the sample must be facing that direction. This is the working principle of MuSR

(b) A scintillating tile schematic. Region A is the scintillator Region B is the cavity that is filled by a transparent resin, green is the wavelength shifting fibre, and the white are the studs (note not to scale)

Figure 3: Schematics for a basic MuSR setup (left) and a example scintillating tile for Super MuSR (right).

## 2.3 Transmittance

One metric being investigated here is transmittance, which is governed by the Beer-Lambert law [12]:

$$T(x, \lambda) = \frac{I}{I_0} \quad ? \quad I = I_0 \exp(-\alpha x) \quad (5)$$

The greater the fraction of light that passes through the epoxy resin samples, the better the tile perform.

## 3 Experimental

### 3.1 Bulk Epoxy Testing

#### 3.1.1 Procedure

One aspect of the tile is a clear epoxy that fills in the gap between the WLSF and the Scintillating Plastic. UV cuvettes were filled with various epoxies to determine the highest bulk transparency using a photo-spectrometer. Once calibrated using distilled water as a baseline, cuvettes were placed into the photo-spectrometer, which determined the transmittance with respect to wavelength of the epoxies. To determine epoxy performance integrated counts is calculated in the wavelength region of 400nm-450nm, as this is the wavelengths of photons emitted by the scintillating plastic.

#### 3.1.2 Results & Discussion

Two candidates were selected for comparison, IN2 and RAL71a resin. A large of 18 samples of both resins were tested, an example of one of each resin is plotted in 4. The mean FOM for the IN2 resin was  $50.1842 \pm 0.1095$ . For the RAL71a, the mean was  $49.9397 \pm 0.1253$ . This showed that the IN2 resin was clearer, in terms of bulk transmittance. IN2 resin was found to have a peculiar "cloudy" surface finish which reduced light output and hence was deemed unsuitable for production, irrespective of the superior bulk transmittance. As a result the RAL71a resin was chosen for the 30mm tiles.

illustration

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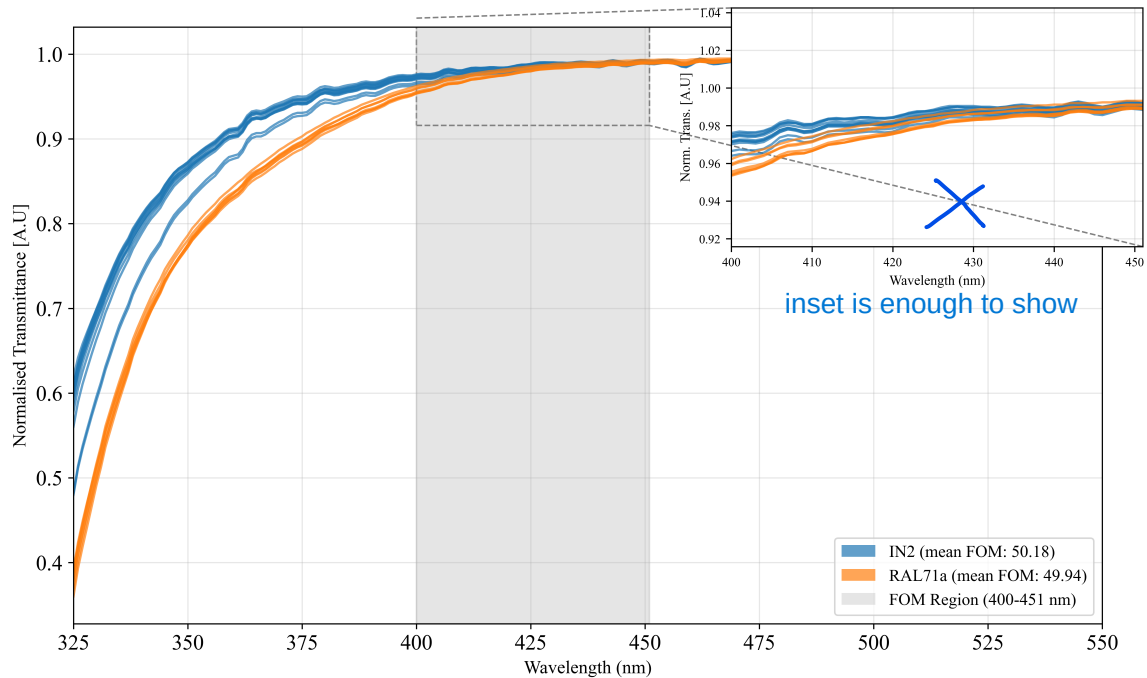


Figure 4: Example Transmittance against Wavelength Plots for a RAL71A and IN2 Resin. Note the grey integrated counts Region.

## 3.2 Scintillating Tile Testing

### 3.2.1 Setup & Calibration

To mimic the incident radiation released by muons, a Strontium-90 source is attached to a clamp and lab jack allowing for safe 3 dimensional movement. The beam is collimated and aligned onto the scintillating tile. The tiles vary in length, ranging from 30mm to 210mm. To accommodate this two PMTs are attached to optical rail to ensure that they are level, and then placed inside a light tight box. Signal from the PMT is sent to a picoscope to be processed into a pulse height spectra (PHS), which is a collection of different pulse amplitudes, plotted against counts, giving an idea of the distribution of the incident radiation [10]. Then the PHS is analysed using code to extract the peak value. Calibration involves gain matching the two PMTs, this is done by placing a benchmark tile on each PMT, and doing a voltage scan. The result is a PHS peak against Voltage plot. The PMTs can then be selected to each have a set benchmark tile peak value, such that the PMTs are calibrated. An example of a calibration scan can be found in figure 5.

### 3.2.2 Procedure

For a given tile, the PMT rig and tile studs are cleaned using a Mix for 70% Isopropanol and 30% deionised water, this is to prevent cracking of the WLSF while ensuring the tiles remain clean. Tiles are coupled to the PMTs using optical grease. Then the source is aligned to the centre of the tile, which is done by marking the middle of the tile and using a reticle designed to replicate the direction the collimated  $^{90}\text{Sr}$  beam will be pointing. The peak value is extracted using the peak finding software (example in figure 6). A benchmark tile that has been tested on the TS1 Beamline, and was determined to have an acceptable light output sets the acceptance threshold. This "Benchmark Tile" is used for comparison to the new production tiles while ensuring no change in PMT behaviour over the days measurements take place. The value of of the benchmark tile was

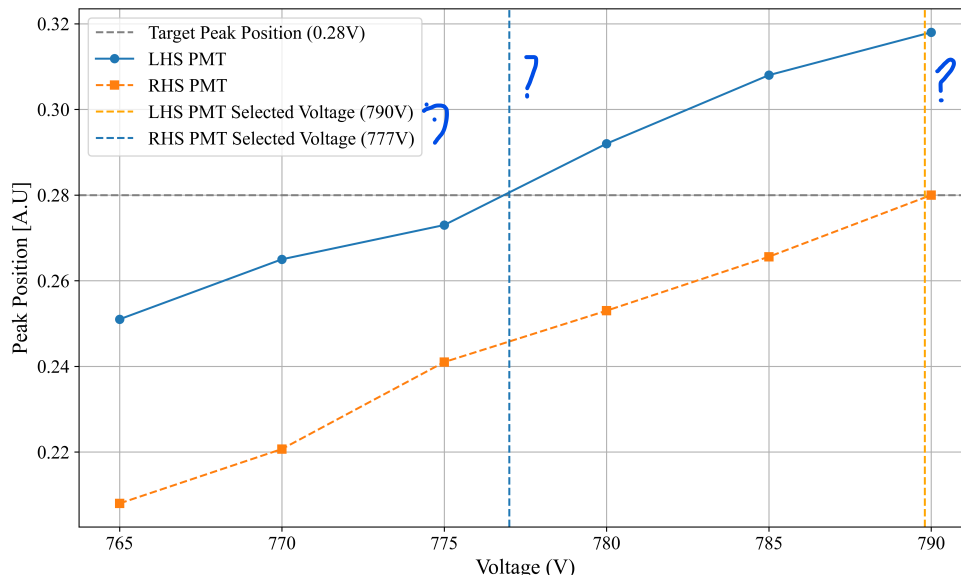


Figure 5: Example of calibration of the left hand side (LHS) and right hand side (RHS) Voltage against PHS peak. For a desired PHS peak position of the benchmark tile of 0.28, the LHS and RHS PMTs were given voltages of 777 and 790 respectively.

set at 0.2592V, which was its best performing run throughout all repeats. This assures relative tile performance is certainly better than the benchmark tile. Poor performing tiles are referenced against the production teams notes to examine where in the defect may have arisen. Two repeats were conducted per tile to ensure good optical coupling. For all tiles the highest value peak was selected because error such as the optical coupling, peak finding and electrical noise can only reduce the peak value. To estimate error, the use of all the benchmark tile repeats were used to get an idea of the standard deviation of the whole system. Determination of errors can be found in the appendix (see A.1).

### 3.2.3 Results & Discussion

Within the scatter plot, there are 4 tiles that are to be investigated, two tiles that the actual peak value falls below the acceptance threshold and 2 with error bars below the threshold. Tiles 43 and 47 were the only tiles that overlapped with the benchmark tile distribution in the histogram. Tile 43 was noted to have "botched studs", and tile 47 was noted to have a myriad of issues, including under-filled and rough resin, bubbles in the epoxy, and other imperfections, issues with the remaining 5 tiles are outlined in table 1. These issues found in tiles 45 and 60 are not significant compared to tiles 47 and 43, with minute defects also present in tiles that perform well. Therefore only the two tiles with actual peak value fell below threshold will be discarded. The histogram provided interesting results, with two prominent peaks. As the data has multiple batches of tiles, and production is an evolving process, changes to processes include new ESR, and new stud lapping jigs (better stud polish leading to greater light output) and improvements to technique can be attributed to a shift in the distribution by improving the light output. Unfortunately there is no clear change in production documented in the production team spreadsheet. In the scatter diagram, there is a clear improvement after tile 76, suggesting some change in production technique has caused this. For the Benchmark tile, there

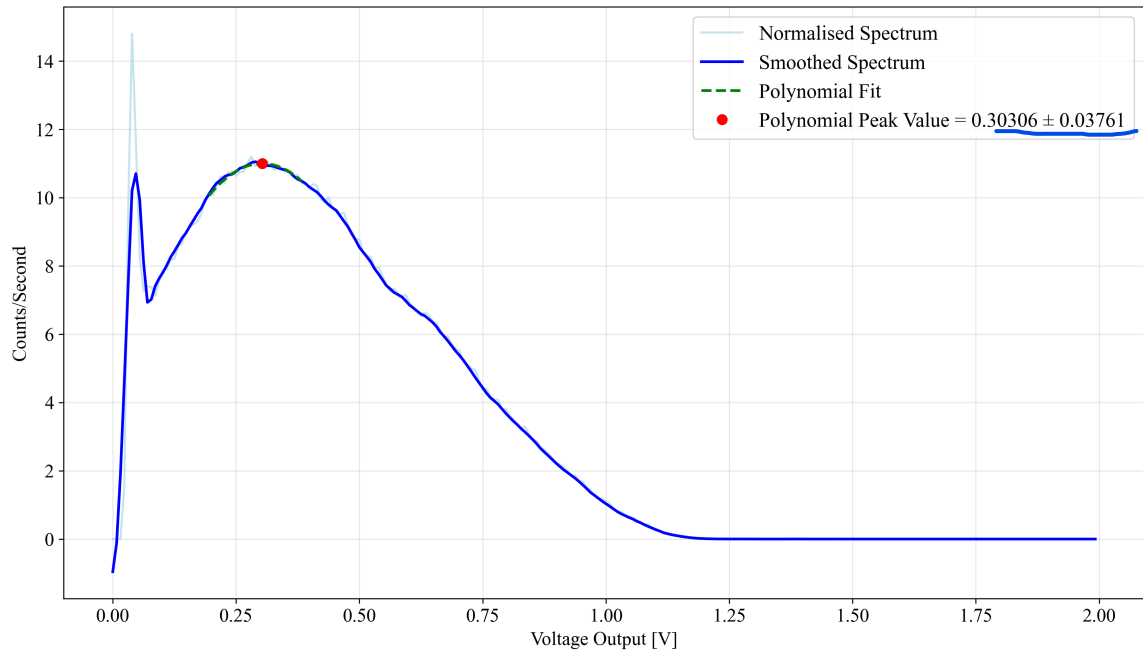


Figure 6: An Example of the peak finder locating peak of the PHS. The Peak is normalised by runtime, and other picoscope data such as trigger level is extracted from the settings file. Polynomial fitting is done using `scipy.optimize.curve_fit`, after an initial peak guess of `scipy.signal.find_peaks`. The plots are smoothed for appearance using a Savitzky-Golay filter [13].

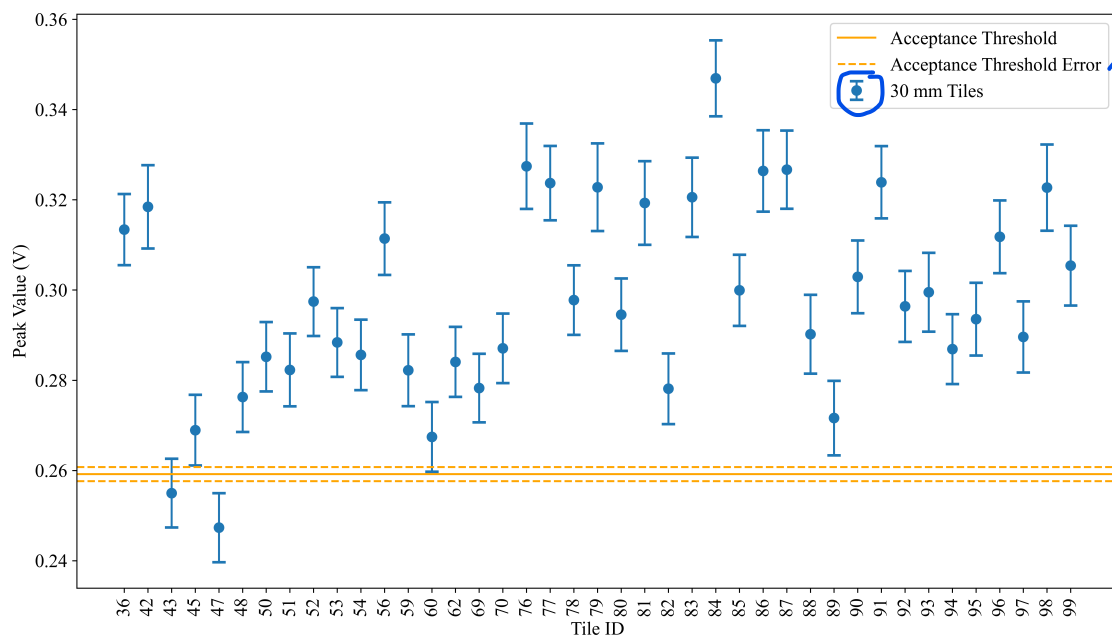


Figure 7: Scatter Graph of tile peak value Against Tile ID.



| Tile ID | Peak Value | Production Notes                                                     |
|---------|------------|----------------------------------------------------------------------|
| 45      | 0.2689     | Small bubbles in leg & stud, possible contamination.                 |
| 60      | 0.2675     | Ultrasonic washed stud, unwashed tile, clear resin.                  |
| 69      | 0.2783     | Tiny hair in one stud                                                |
| 76*     | 0.3163     | Tiny hair in one stud, tiny bubble in each stud, divots slightly low |
| 82      | 0.2781     | Single tiny bubble in leg & stud, divots topped up with resin        |
| 89      | 0.2716     | Tiny bubble in one stud, divots low                                  |
| 98*     | 0.3227     | Bubble in one leg, tiny bubble in one stud                           |

Table 1: Summarised notes from the production team spreadsheet for a range of tile performances. Also included at are various tiles that performed well, which are marked with a "\*" to provide an idea of the comments for all ranges of tile performance. Note: when referring to divots, this is the concave depression in the tiles that are around the region where the WLSF exits the scintillating plastic.

was a mean peak value of  $0.2462 \pm 0.0820V$ , compared to a value of  $0.2978 \pm 0.0216V$  for the 41 tiles being tested. This gives the new batch of tiles a 20.9% average improvement compared to the benchmark tile. To determine an overall expected percentage of tiles produced, approximately 2.7% of tiles are broken before they are tested, and of that remaining fraction, approximately 4.9% of tiles do not pass. Therefore the total projected percentage of tiles that reach the final detector assembly stage is 92.5%.

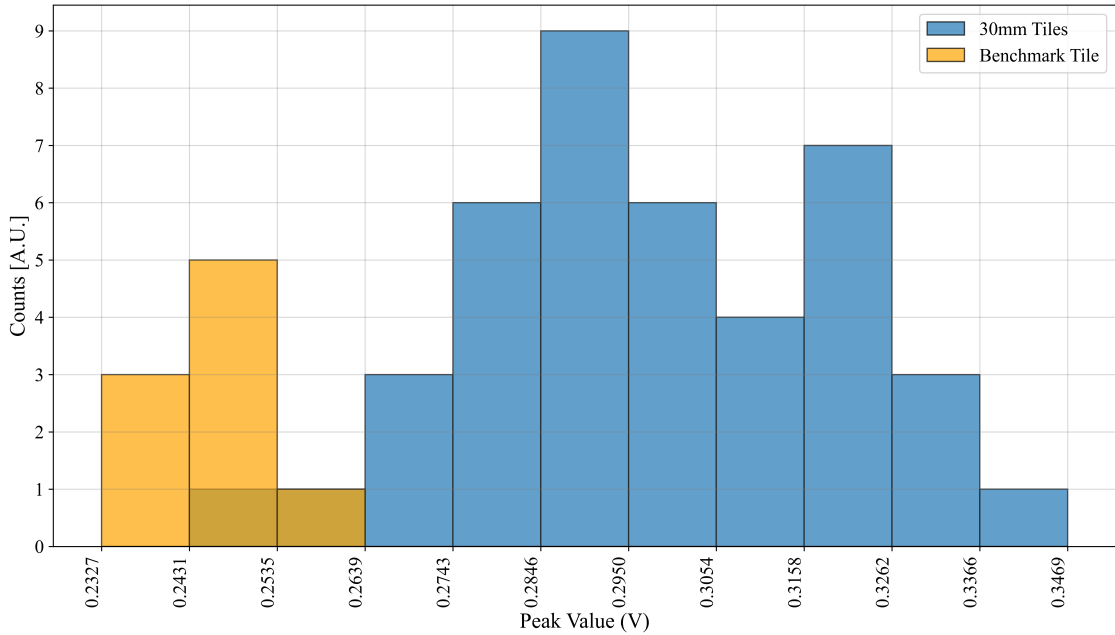


Figure 8: Histogram of all 41 Tiles selected for testing, and the Benchmark tile repeats.

## 4 Conclusion & Next Steps

### 4.1 Bulk Epoxy Testing

In conclusion RAL71a resin is suitable for 30mm, 43mm and 63mm tiles, despite having worse bulk transmittance. However when it is being used for longer length tiles, such as the warping of tiles when curing, as a result other epoxy resins are being tested to allow for a resin to be chosen for longer length tiles. Further work will involve the testing and investigation of different epoxies for the longer length tiles. A method of extracting raw intensity and error needs to be determined, as the photo-spectrometer only returns relative intensity compared to an arbitrary baseline, in this case distilled water.

are the methods to reduce the stress without changing material?

### 4.2 Scintillating Tile Testing

For the batch of 30mm tiles, it was found that going forward the expected percentage of tiles that reach the detector assembly stage is 92.5%. The current tile testing setup is suitable for determining acceptance threshold for all tiles, future work will involve producing similar databases for all lengths of tile, selecting a benchmark tile for each length which will be done during the February 2026 cycle at ISIS. New tiles coming of the production have a mean improvement of 20.9%, with mean peak values of  $0.2462 \pm 0.0820V$  and  $0.2978 \pm 0.0216V$  for old and new tiles respectively. Over time the tile testing error threshold aims to move away from a single benchmark tile, as damage and wear on the tile could be catastrophic for determining performance thresholds, benchmark tiles will be measured over time to determine chanted to the tile or the experimental setup. In conclusion, the production team should continue production with no systematic defects present within the 30mm tiles, and to reach the 440 total 30mm tiles required should aim to produce 476 tiles to account for breakages and poor light output, however another batch will be tested at a later date to confirm that the percentage of tiles reaching the assembly stage is maintained or improved.

Overall this is a good start. The structure was good and report had technical details. Main things to look for:

Better explanation of the physics super MuSR. Especially what field it measures, where and what for. You can give some graphs from papers with typical results and ideally correlate the proposed improvements with science.

proofing of the text, too many inconsistencies (Capitals!) and even missing words.

More illustrations (like you've done in the presentation). Especially of the apparatus, rig, etc. Diagrams sometimes better than photos, as they are not cluttered, but photos like you did of a tile - are excellent.

For future plans, review what has been done before on the epoxy and search for any other examples of epoxy cracking, curing etc. There might be similar issues somewhere else (LHC, LIGO, etc) where it was addressed.

Less notebook style of writing, first we done this, then that, etc. More like a scientific paper/report

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## A Appendix

### A.1 Error Estimation

#### A.1.1 Epoxy Testing

For the epoxy testing, the resolution of the photo-spectrometer is 0.0005, hence the error is dominated by the SEM on the repeats [14]:

$$SEM = \frac{\sigma_{exp}}{\sqrt{N}} \quad (6)$$

Where  $N$  is the number of repeats, and  $\sigma$  is the sample standard deviation, which for  $N$  measurements  $x_1, x_2, \dots, x_N$  is given by [14]:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (7)$$

### A.1.2 Tile Testing

There are two sources of error, ones contributing to the error from the experimental setup, such as electronics, counting statistics and grease coupling, the other is from the error on the peak finding code. To determine errors on the individual tiles, the error on the standard deviation from the whole system and the peak finding code was added in quadrature, of which the experimental setup dominated. typical values were 0.00218 and 0.00751 for the peak finder and experimental setup respectively. Error for tile  $i$  is given by:

$$\sqrt{(\sigma_p)^2 + (\sigma_{exp})^2} \quad (8)$$

where  $\sigma_p$  is the standard deviation in the peak-finding software, and  $\sigma_{exp}$  is the standard deviation in the experimental setup, determined by the benchmark tile repeats. Error on the acceptance threshold was also calculated using the Standard Error on the Mean (SEM) [14] on the peak value, with many repeats.

good use  
of appendix